Supplementary Material
Sparse Counterfactual Attribution of Crop-Yield Anomalies

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1 Driver Features and Robustness Tables

Table 1: Full feature list for each grouped-SCAA driver group.

driver group	features
heat	heat_days_30, heat_days_35, heatwave_events_3d_30, heat_degree_days_30, season_tmax_mean
drought	rain_sum, rain_mean, dry_days_1mm, max_dry_spell_1mm, dry_spell_events_7d, dry_spell_events_14d, hot_dry_days_30_1mm
frost_cold	frost_days_0, cold_days_5, min_tmin, frost_events_2d, season_tmin_mean
excess_rain	wet_days_1mm, heavy_rain_days_10, heavy_rain_days_20, max_1day_rain, max_3day_rain, max_7day_rain
radiation	radiation_sum, radiation_mean

Table 2: Sensitivity of leave-one-event-year-out SCAA summaries to the anomaly z-threshold.

threshold	anomaly_count	share_of_dataset	median_recoverable_fraction	top_driver_groups
z < -1.0	214	0.17	0.116	drought, heat, excess_rain
z < -1.5	85	0.068	0.121	drought, heat, excess_rain
z < -2.0	30	0.024	0.119	drought, heat, radiation

Table 3: Robustness of low-yield anomaly membership to detrending choice.

detrending_method	anomaly_count	overlap_with_linear	jaccard_with_linear
linear	214	214	1.0
quadratic	194	178	0.774
centered_7yr_rolling	186	142	0.55

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Table 4: Observed crop-state support for vulnerability profiles.

crop	crop-state pairs	rows	states
Barley	10	283	Colorado, Kansas, Minnesota, Montana, Nebraska, North Dakota, Oklahoma, South Dakota, Texas, Washington
Canola	6	133	Kansas, Minnesota, Montana, North Dakota, Oklahoma, Washington
Oats	12	416	Colorado, Illinois, Iowa, Kansas, Minnesota, Montana, Nebraska, North Dakota, Oklahoma, South Dakota, Texas, Washington
Wheat	12	425	Colorado, Illinois, Iowa, Kansas, Minnesota, Montana, Nebraska, North Dakota, Oklahoma, South Dakota, Texas, Washington

2 Event-Year Consistency Details

Table 5: External evidence used to pre-specify expected event-year stress groups.

year	expected group	affected scope	external source	pre-specified
2012	heat, drought	Central United States and Plains crop belt; crop-state rows in the processed frame	NOAA/NCEI Annual 2012 Drought Report; NWS 2012 drought summary	yes
2021	heat, drought	Northern Plains and Canadian Prairie drought context; U.S. crop-state rows in the processed frame	Drought.gov/NIDIS report on the 2020-2021 Northern Plains and Canadian Prairies drought	yes
2022	heat, drought, excess_rain	U.S. drought and regional wetness episodes during the 2022 growing season	NOAA/NCEI Annual 2022 and August 2022 Drought Reports	yes

Table 6: Leave-one-event-year-out event-year consistency summary for 2012, 2021, and 2022.

method	year	n_events	expected_match_rate	median_recoverable
Leave-one-event-year-out grouped SCAA	2012	3	1.0	0.488
Leave-one-event-year-out grouped SCAA	2021	19	1.0	0.18
Leave-one-event-year-out grouped SCAA	2022	13	0.769	0.182

Table 7: Null baselines for event-year consistency checks.

Method	Expected match rate	Median recovery	n event r
Always drought	1.000	–	35
Always heat	1.000	–	35
Most frequent SCAA driver among 2012/2021/2022 event rows (heat)	1.000	–	35
Driver-frequency random	0.896	–	35
Leave-one-event-year-out grouped SCAA	0.914	0.182	35

Note: the most-frequent-driver baseline is computed only on the 35 event-year rows used in the consistency check, not on the full low-yield anomaly set.

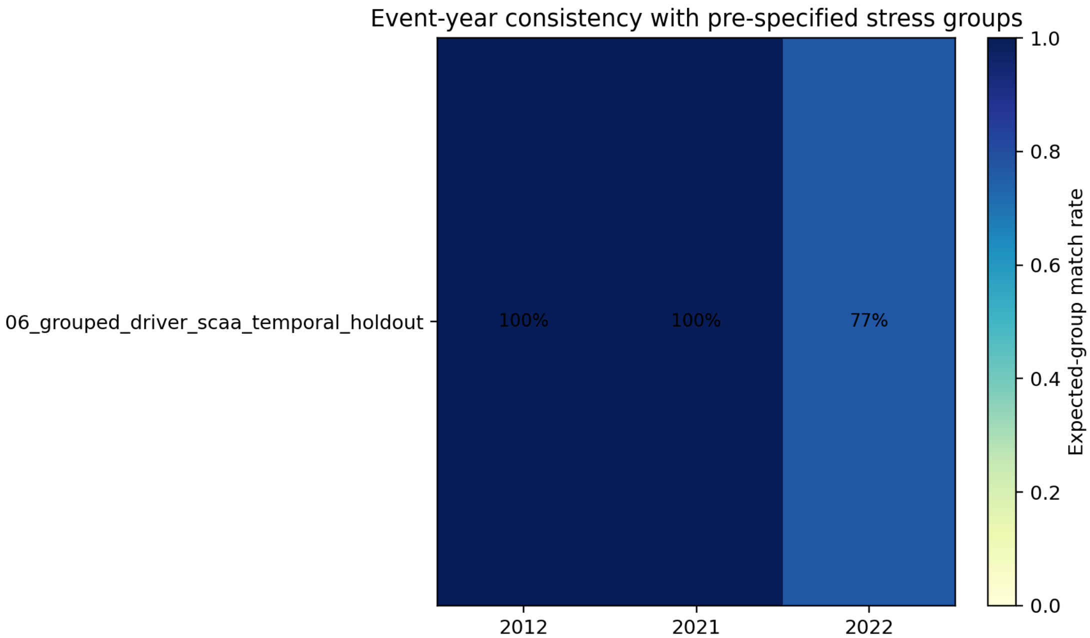

Figure 1: Event-year consistency with pre-specified heat, drought, and moisture stress groups.

3 Early-Warning Extension

This section reports the exploratory early-warning metrics referenced in the main manuscript.

Table 8: Numeric early-warning performance for the exploratory early-warning extension.

stage	ROC AUC	AP	Brier	Top 10%	Top 20%	Threshold
Early-season	0.473	0.225	0.216	0.208	0.229	0.14
Early + mid-season	0.597	0.301	0.185	0.333	0.333	0.16

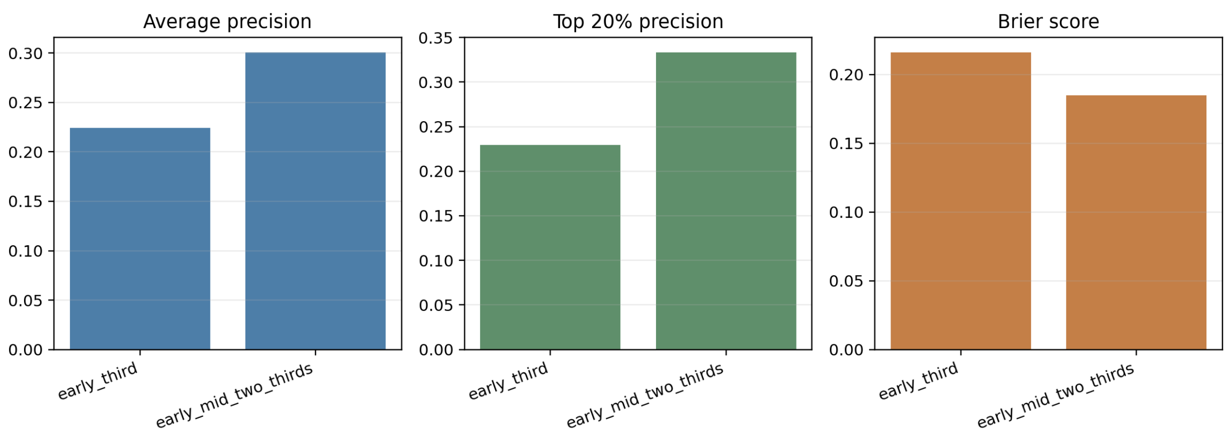

Figure 2: Early-warning extension comparing early-season and early-plus-mid-season anomaly risk models.

4 Reference Mapping

This section maps the cited references to their roles in the manuscript.

Table 9: Reference mapping to manuscript sections.

paper section	use	bib keys
Introduction	Extreme weather and crop-yield losses	Lesk2016; Zampieri2017; Vogel2019; Ray2015
Related work	Machine-learning crop-yield prediction	Paudel2021; Meroni2021; Khaki2019; Leng-Hall2020
Related work	Yield anomalies and detrending	Lu2017; Ray2015; Meng2024; Sjulgard2023
Related work	Interpretable ML for yield models	LundbergLee2017; Ribeiro2016
Method	Sparse and feasible counterfactual explanation	Wachter2018; Mothilal2020; Ustun2019; Poyi-adzi2020; Verma2024
Method and limitations	Event-attribution language and pitfalls	Hannart2016; Otto2017; Oldenborgh2021
Data	Weather and yield data sources	NASAPOWER2025; USDANASSQuickStats
Extension	Early warning and conformal uncertainty	Anderson2024; Meroni2021; Singh2024; Farag2025